

DOCKER COMPOSE

Docker Compose is a tool that allows you to define and manage multi-container Docker applications. It simplifies the process of running multiple containers, their configurations, and their interdependencies. Compose uses a YAML file to define the services, networks, and volumes required for your application.

- ❓ Docker Compose is a tool which is used to manage multi container-based applications.
- ❓ Using Docker Compose we can easily setup & deploy multi container-based applications.
- ❓ We will give containers information to Docker Compose using YML file (docker-compose.yml)
- ❓ Docker Compose YML should have all the information related to containers creation.
- ❓ Docker Compose YML File Looks Like:

#DOCKER COMMANDS FOR UBUNTU

```
$ sudo apt update -y
```

```
$ sudo apt install docker -y
```

```
$ sudo service docker start (or) sudo systemctl start docker
```

```
$ sudo service docker enable (or) sudo systemctl enable docker
```

JENKINS

Jenkins is an open-source automation tool written in Java programming language that allows continuous integration. Jenkins offers a straightforward way to set up a continuous integration or continuous delivery environment for almost any combination of languages and source code repositories using pipelines, as well as automating other routine development tasks.

The following are the main or most popular Jenkins use cases:

- Continuous Integration: With Jenkins pipelines, we can achieve CI for both applications and infrastructure as code.
- Continuous Delivery: You can set up well-defined and automated application delivery workflows with Jenkins pipelines.

Jenkins achieves CI (Continuous Integration) and CD (Continuous Deployment) with the help of plugins. Plugins are used to allow the integration of various DevOps stages. If you want to integrate a particular tool, you must install the plugins for that tool.

ADVANTAGES OF JENKINS:

- It is an open-source tool.
- It is free of cost.
- It does not require additional installations or components. Means it is easy to install.
- Easily configurable.
- It supports 1000 or more plugins to ease your work. If a plugin does not exist, you can write the script for it and share with community.
- It is built in java and hence it is portable.
- It is platform independent. It is available for all platforms and different operating systems. Like OS X, Windows, or Linux.
- Easy support since its open source and widely used.
- Jenkins also supports cloud-based architecture so that we can deploy Jenkins in cloud-based platforms.

```
pipeline {  
    agent any
```

```
    stages {  
        stage('SCM') {  
            steps {  
                git branch: 'main', url: 'https://github.com/Ashwanth3637/devops-guvi.git'  
            }  
        }  
        stage('Built') {  
            steps {  
                bat 'mvn clean install'
```


Inbox (2,859) - ashwanth.22c... x p1 #3 [Jenkins] x Ashwanth3637 (Ashwanth S) x + -

localhost:8080/job/p1/3/

Jenkins

Ashwanth S log out

Dashboard > p1 > #3

Status

Changes

Console Output

Edit Build Information

Delete build '#3'

Timings

Git Build Data

Pipeline Overview

Pipeline Console

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Pipeline Steps

Workspaces

✓ #3 (Mar 19, 2025, 4:57:47 PM)

Add description Keep this build forever

Started by user Ashwanth S

Started 10 hr ago
Took 30 sec

This run spent:

- 14 ms waiting;
- 30 sec build duration;
- 30 sec total from scheduled to completion.

Revision: 70ad7e6e58ba0f8c9d4c1d39ac550d7b7edc915a
Repository: <https://github.com/Ashwanth3637/Jenkins.git>

- refs/remotes/origin/master

No changes.

localhost:8080/job/p1/3/console

Dashboard > p1 > #3

Changes

Console Output

Edit Build Information

Delete build '#3'

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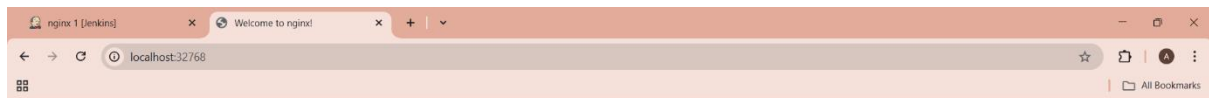
Workspaces

Previous Build

```
Started by user Ashwanth S
Obtained jenkinsfile from git https://github.com/Ashwanth3637/Jenkins.git
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in /var/lib/jenkins/workspace/p1
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Declarative: Checkout SCM)
[Pipeline] checkout
Selected Git installation does not exist. Using Default
The recommended git tool is: NONE
using credential token
> git rev-parse --resolve-git-dir /var/lib/jenkins/workspace/p1/.git # timeout=10
Fetching changes from the remote Git repository
> git config remote.origin.url https://github.com/Ashwanth3637/Jenkins.git # timeout=10
Fetching upstream changes from https://github.com/Ashwanth3637/Jenkins.git
> git --version # timeout=10
> git --version # 'git version 2.43.0'
using GIT_ASKPASS to set credentials token
> git fetch --tags --force --progress -- https://github.com/Ashwanth3637/Jenkins.git +refs/heads/*:refs/remotes/origin/* #
timeout=10
> git rev-parse refs/remotes/origin/master^{commit} # timeout=10
Checking out Revision 70ad7e6e58ba0f8c9d4c1d39ac550d7b7edc915a (refs/remotes/origin/master)
> git config core.sparsecheckout # timeout=10
```

09:16 20-03-2025

09:17 20-03-2025

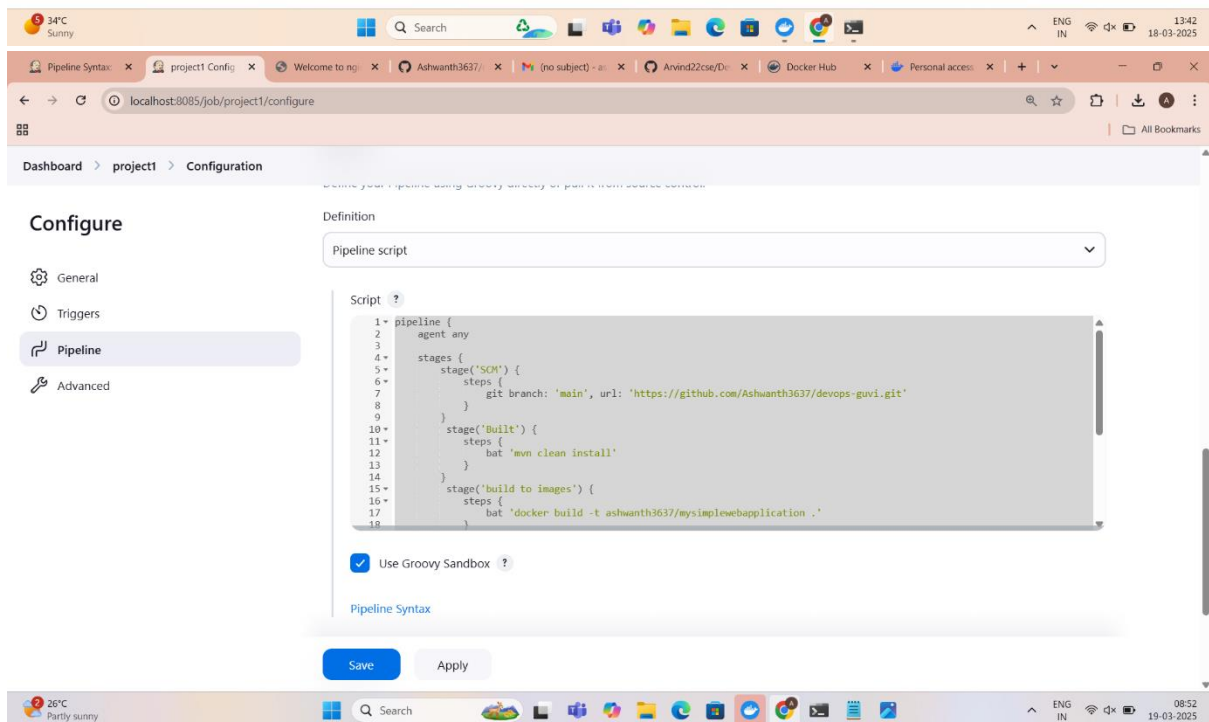


Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.



The screenshot shows the Jenkins web interface. The top navigation bar includes 'Dashboard', 'project1', and 'Configuration'. The main heading is 'Configure'. On the left, there are tabs for 'General', 'Triggers', 'Pipeline', and 'Advanced'. The 'Pipeline' tab is selected. The 'Definition' section shows 'Pipeline script' as the selected option. Below this, a 'Script' section contains a Groovy script for a pipeline. The script defines a stage 'build to images' with a step to build the application, and another stage 'docker push hub' with a step to push the image to Docker Hub. A checkbox 'Use Groovy Sandbox' is checked. At the bottom, there are 'Save' and 'Apply' buttons.

```
13 }
14 }
15 stage('build to images') {
16   steps {
17     bat 'docker build -t ashwanth3637/mysimplewebapplication .'
18   }
19 }
20 stage('docker push hub'){
21   steps{
22     script{
23       // This step should not normally be used in your script. Consult the inline help for details.
24       withDockerRegistry(credentialsId: 'docker_cred', url: 'https://index.docker.io/v1/') {
25         bat 'docker push ashwanth3637/mysimplewebapplication'
26       }
27     }
28   }
29 }
30 }
```

Dashboard > project1 > Pipeline Syntax

Examples Reference

IntelliJ IDEA GDSDL

withDockerRegistry: Sets up Docker registry endpoint

withDockerRegistry ?

Docker registry URL ?

https://index.docker.io/v1/

URL to the Docker registry you are using. May be left blank to use the public DockerHub registry (currently https://index.docker.io/v1/). (from Docker Commons Plugin)

Registry credentials

ashwanth3637/*****

+ Add

Generate Pipeline Script

// This step should not normally be used in your script. Consult the inline help for details.
withDockerRegistry(credentialsId: 'docker_cred', url: 'https://index.docker.io/v1/') {
 // some block
}

Jenkins

Search

Ashwanth S

log out

Dashboard > project1 > #6

Status

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Thread Dump

Pause/resume

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Pipeline Steps

#6 (Mar 18, 2025, 4:28:25 PM)

Add description

Keep this build forever

Started by user Ashwanth S

Started 3 sec ago

Build has been executing for 3 sec

This run spent 7 ms waiting in the queue.

Revision: 803e5bc485a8f3dc636a0b40d05f35c9cd96ba47

Repository: https://github.com/Ashwanth3637/devops-guvi.git

refs/remotes/origin/main

No changes.

Pipeline Syntax X project1 #6 [jenkins] X Welcome to Jenkins X Ashwanth3637/ X (no subject) - ashwanth3637 X Arvind22cse/Docker Hub X Personal access token X

localhost:8085/job/project1/6/

All Bookmarks

 **Jenkins**

Search 🔍 Notifications 🔔 Security 🔒 User: Ashwanth S Logout

Dashboard > project1 > #6

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✔ #6 (Mar 18, 2025, 4:28:25 PM)

🕒

Started by user [Ashwanth S](#)

Started 3 sec ago
Build has been executing for 3 sec

🕒

This run spent 7 ms waiting in the queue.

git

Revision: 803e5bc485a8f3dc636a0b40d05f35c9cd96ba47
Repository: <https://github.com/Ashwanth3637/devops-guvf.git>

- refs/remotes/origin/main

</>

No changes.

Add description

Keep this build forever

36°C Sunny

Search

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16:34 18-03-2025